Topology: MAT307/M405/MAT803 : - Problem Set 4

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1. Let X and Y be two topological spaces; and $a \in X$, $b \in Y$ be two given points. Prove that the maps $f : X \to X \times Y$ and $g : Y \to X \times Y$ given by

$$f(x) = (x,\ b) \quad \text{and} \quad g(y) = (a,\ y)$$

are embeddings.

2. Suppose that $f,\ g : X \to \mathbb{R}$ are two continuous maps.
   (i). Prove that the set $\{x \in X \mid f(x) \leq g(x)\}$ is closed in X.
   (ii). Prove that the map $h : X \to \mathbb{R}$ given by $h(x) = \min\{f(x),\ g(x)\}$ is continuous.

3. Let $\{A_\alpha\}$ be a collection of subsets of X such that $X = \cup_\alpha A_\alpha$. Suppose that $f : X \to Y$ is a map such that the restriction map $f_{A_\alpha} : A_\alpha \to Y$ is continuous for each α.
   (i). Prove that if the collection $\{A_\alpha\}$ is finite and each set A_α is closed, then f is continuous.
   (ii). Find an example where the collection is $\{A_\alpha\}$ is countable and each set A_α is closed, but f is not continuous.
   (iii). An indexed family of sets $\{A_\alpha\}$ is said to be **locally finite** if each point x has a neighborhood that intersects $\{A_\alpha\}$ for finitely many values of α. Prove that if the family $\{A_\alpha\}$ is locally finite and each set A_α is closed, then f is continuous.

4. Let $f : X \to A$ and $g : Y \to B$ be two continuous maps. Define $f \times g : X \times Y \to A \times B$ by $(f \times g)(x,\ y) = (f(x),\ g(y))$. Prove that $f \times g$ is a continuous map.

5. Let (X, d) be a metric space and let $S \subseteq X$ be a non-empty subset. For any $x \in X$, define the distance from x to S by $d(x, S) = \inf_{s \in S} d(x, s)$.
   (1) Show that for any $x, y \in X$, $|d(x, S) - d(y, S)| \leq d(x, y)$.
   (2) Conclude that $f_S : X \to \mathbb{R}$ defined by $f_S(x) = d(x, S)$ is continuous on X.

6. Let $\{A_n\}_{n \in \mathbb{N}}$ be a sequence of connected subspaces of X such that $A_n \cap A_{n+1} \neq \emptyset$ for all $n \in \mathbb{N}$. Prove that $\bigcup_{n=1}^{\infty} A_n$ is connected.

7. Determine whether or not $\mathbb{R}^\omega$ is connected under the uniform topology.

8. Determine whether or not the lower limit topology $\mathbb{R}_\ell$ on $\mathbb{R}$ is connected.

9. Let $C \subset X$ be a connected subspace of X and $A \subset X$. Prove that if C intersects both A and $X \setminus A$, then C intersects Bd A.

Tushar Kanta Naik, Reader F, SMS, NISER

10. Let X and Y be two connected subspaces and $A \subsetneq X$ and $B \subsetneq Y$. Prove that $(X \times Y) \setminus (A \times B)$ is connected.

11. Let $Y \subset X$; and let X and Y be connected. Prove that if A and B forms a separation of $X \setminus Y$, then both $Y \cup A$ and $Y \cup B$ are connected.

Tushar Kanta Naik, Reader F, SMS, NISER